

RaceDay System — Section B: API Endpoint Plan

Purpose

This document defines the planned API endpoints for the RaceDay system before implementation. The plan covers authentication, user profiles, events, categories, event enrolments, and race results.

API Endpoint Table

Coverage of Required Functionality

Implementation Note

The implemented API in Part 2 should closely follow this endpoint plan. Any changes made during implementation should be documented and explained in the README.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account in the RaceDay system.	None (public)	firstName, lastName, email, password, phone, role	201 Created – User registered successfully. 409 Conflict – Email already exists. 400 Bad Request – Invalid data.
POST	/api/auth/login	Authenticates a user using their email and password.	None (public)	email, password	200 OK – Authentication token and user details. 401 Unauthorized – Invalid credentials.
GET	/api/profile	Retrieves the profile of the currently logged-in user.	Any (logged in)	None	200 OK – User profile. 401 Unauthorized – User is not logged in.
PUT	/api/profile	Updates the profile information of the logged-in user.	Any (logged in)	firstName, lastName, phone	200 OK – Updated profile. 400 Bad Request – Invalid data. 401 Unauthorized – Not logged in.
GET	/api/events	Retrieves a list of all available race events.	None (public)	None	200 OK – List of events.
GET	/api/events/{id}	Retrieves detailed information about a specific race event.	None (public)	None	200 OK – Event details. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new race event.	Organiser	name, description, eventDate, location	201 Created – Event created. 400 Bad Request – Invalid data. 401 Unauthorized – Not logged in.
PUT	/api/events/{id}	Updates an existing race event.	Organiser	name, description, eventDate, location	200 OK – Event updated. 404 Not Found – Event does not exist.
DELETE	/api/events/{id}	Deletes an existing race event.	Organiser	None	204 No Content – Event deleted. 404 Not Found – Event does not exist.
GET	/api/categories	Retrieves all race categories available in the system.	None (public)	None	200 OK – List of categories.
GET	/api/categories/{id}	Retrieves details of a specific race category.	None (public)	None	200 OK – Category details. 404 Not Found – Category does not exist.
POST	/api/categories	Creates a new race category.	Organiser	name, description	201 Created – Category created. 400 Bad Request – Invalid data.
PUT	/api/categories/{id}	Updates an existing race category.	Organiser	name, description	200 OK – Category updated. 404 Not Found – Category does not exist.
DELETE	/api/categories/{id}	Removes a race category from the system.	Organiser	None	204 No Content – Category deleted. 404 Not Found – Category does not exist.
GET	/api/events/{eventId}/categories	Retrieves all categories associated with a specific event.	None (public)	None	200 OK – List of event categories. 404 Not Found – Event does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/categories	Associates a category with a race event.	Organiser	categoryId	201 Created – Category assigned. 404 Not Found – Event/category does not exist. 409 Conflict – Category already assigned.
POST	/api/enrolments	Enrols the logged-in participant in a race event and category.	Participant	eventId, categoryId	201 Created – Enrolment created. 404 Not Found – Event/category does not exist. 409 Conflict – Already enrolled.
GET	/api/enrolments	Retrieves all enrolments belonging to the logged-in participant.	Participant	None	200 OK – List of enrolments. 401 Unauthorized – Not logged in.
GET	/api/events/{eventId}/enrolments	Retrieves participants enrolled in a specific race event.	Organiser	None	200 OK – List of enrolments. 404 Not Found – Event does not exist.
GET	/api/enrolments/{id}	Retrieves details of a specific enrolment.	Participant	None	200 OK – Enrolment details. 404 Not Found – Enrolment does not exist.
DELETE	/api/enrolments/{id}	Cancels an existing participant enrolment.	Participant	None	204 No Content – Enrolment cancelled. 404 Not Found – Enrolment does not exist.
GET	/api/results	Retrieves all recorded race results.	None (public)	None	200 OK – List of results.
GET	/api/results/{id}	Retrieves a specific race result.	None (public)	None	200 OK – Result details. 404 Not Found – Result does not exist.
GET	/api/events/{eventId}/results	Retrieves all results for a specific race event.	None (public)	None	200 OK – Event results. 404 Not Found – Event does not exist.
POST	/api/results	Records a race result for an enrolled participant.	Organiser	enrolmentId, finishTime, position	201 Created – Result recorded. 400 Bad Request – Invalid data. 404 Not Found – Enrolment does not exist.
PUT	/api/results/{id}	Updates an existing race result.	Organiser	finishTime, position	200 OK – Result updated. 404 Not Found – Result does not exist.
DELETE	/api/results/{id}	Deletes a race result.	Organiser	None	204 No Content – Result deleted. 404 Not Found – Result does not exist.

Required Area	Covered
Authentication — Register	Yes
Authentication — Login	Yes
User Profile	Yes
Events	Yes
Categories	Yes
Event Enrolments	Yes
Results	Yes